

Voice Activity Detection for Conversational AI in Clinical Simulation-Based Learning: An Empirical Optimisation Framework

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Abstract—Conversational AI systems are increasingly used as learning technologies in clinical simulation-based education, where interaction quality directly influences measurable learning outcomes. Voice Activity Detection (VAD) in large-language-model-driven speech-to-speech systems is governed by three configurable parameters: activation threshold, silence duration, and prefix padding.

This work presents a two-stage empirical methodology applied to clinical simulation training: (1) a Central Composite Design (CCD) coarse sweep to map the parameter response surface, and (2) Response Surface Methodology (RSM) statistical modelling with Bayesian refinement to identify Pareto-optimal trade-offs between false positive rate and turn latency. Directions for a future adaptive run-time extension are outlined.

Experiments were conducted on a purpose-built Raspberry Pi-based audio terminal running the OpenAI Realtime API, using a standardized OSCE-format consultation script with controlled backchannel vocalizations. Results show that the three-parameter space contains an identifiable Pareto front and that `silence_duration_ms` is the strongest statistically significant predictor of interruptive false positive events, while turn latency is only weakly explained by VAD configuration.

The proposed methodology and open-source implementation provide a transferable framework for empirical optimisation of server-VAD-configurable speech interfaces under comparable deployment conditions.

Index Terms—Learning Technologies, Simulation-Based Learning, Clinical Simulation, Conversational AI in Education, Voice Activity Detection, Bayesian Optimisation, Response Surface Methodology, Human-AI Interaction.

1 INTRODUCTION

THE emergence of large language model (LLM)-driven speech-to-speech systems has introduced a new class of conversational interfaces with direct applications in health-care education, clinical simulation, and professional skills training. Systems of this kind rely on Voice Activity Detection (VAD) to segment continuous audio into discrete speaker turns, a process that fundamentally shapes the perceived naturalness of the interaction.

Turn-taking in spoken interaction is a tightly timed process, with typical response gaps on the order of a few hundred milliseconds, making conversational systems highly sensitive to turn detection errors [1].

When VAD is misconfigured, two failure modes emerge: *false positive triggers*; where backchannel vocalizations such

as affirmative sounds or brief pauses are misidentified as new speaker turns; and *missed turns*, where genuine speech onset goes undetected due to excessive sensitivity thresholds. Three parameters govern server-side VAD behavior in contemporary implementations: activation threshold, silence duration, and prefix padding.

Despite their obvious influence on conversational quality, no published empirical framework maps their joint effects on turn detection performance in educational or clinical simulation contexts. Practitioner guidance remains largely informal, with limited systematic evaluation in the literature [2], [3]. This absence is consequential: in Objective Structured Clinical Examination (OSCE) scenarios, where trainee clinicians engage with AI-simulated patients, unintended interruptions disrupt the ecological validity of the encounter and may bias evaluation outcomes and reduce the reliability of measurable learning assessment.

2 RELATED WORK

2.1 Voice Activity Detection in Educational Technology

VAD has been studied extensively in telephony and automatic speech recognition (ASR) contexts, where the dominant concern is noise robustness rather than conversational naturalness. Classical approaches based on energy thresholding [3], [4] and more recent neural methods [5] have been benchmarked primarily on telephony corpora (*telephone-based conversational speech*) that do not reflect the specific characteristics of clinical simulation dialogues, including deliberate pauses, affirmative backchannels, and the irregular prosody of trainee speakers under assessment pressure.

Within educational technology, VAD is typically treated as an infrastructure component rather than an experimental variable. Intelligent Tutoring Systems (ITS) and dialogue-based learning environments acknowledge VAD as a source of error [6] but do not systematically optimise its parameters for domain-specific conversational patterns. Despite extensive work on VAD in speech processing, most studies optimise detection accuracy under noise conditions rather than conversational interaction quality, leaving a gap between signal-level optimisation and dialogue-level performance [2], [3].

2.2 LLM-Driven Speech Interfaces

OpenAI introduced the GPT-4o omni model with native end-to-end audio processing, reporting average speech response latencies of 320ms comparable to human conversational response times [7]. The subsequent Realtime API, released in public beta in October 2024 [8], exposed this capability as a configurable WebSocket interface for developers.

Neither the system card nor the API documentation provides empirical guidance for VAD parameter configuration. The parameters `threshold`, `silence_duration_ms`, and `prefix_padding_ms` are given as fixed defaults, without any analysis of their individual roles or their combined impact on conversational dynamics.

Related open-source frameworks such as Pipecat [9] document analogous parameters and acknowledge sensitivity to background noise and backchannel vocalisations, yet do not provide any systematic optimisation methodology. The absence of empirical guidance in the primary source documentation itself motivates the present work, particularly given prior findings that turn-taking behaviour in conversational systems is highly sensitive to timing and interaction dynamics [2].

Although the internal implementation of the OpenAI server-side VAD is not publicly documented, inspection of open-source alternatives confirms that the three exposed parameters correspond to canonical components of frame-based energy detection systems. Pipecat’s open-source implementation [9] exposes four parameters with explicit defaults: `confidence` (activation threshold, default 0.7), `stop_secs` (silence before turn end, default 0.8s), `start_secs` (speech duration before onset confirmed, default 0.2s), and `min_volume` (minimum audio level, default 0.6). The first three map directly to the OpenAI parameters `threshold`, `silence_duration_ms`, and `prefix_padding_ms` respectively, indicating that these are standard turn-detection primitives rather than API-specific constructs. Notably, `min_volume` has no exposed equivalent in the OpenAI API, introducing an additional non-controllable factor. This supports treating the OpenAI server-side VAD as a black box with interpretable control dimensions, making response surface modelling an appropriate method for characterising its behaviour with partially observable controlled input conditions.

A semantic VAD variant, which uses a classifier to estimate turn completion from linguistic content rather than silence duration alone, has since been introduced as a default mode in some implementations [10]. While promising, this approach lacks the configurability and interpretability required for domain-specific optimization and is not available in all deployment contexts, including the hardware-constrained terminal studied here.

2.3 Response Surface Methodology in HCI and Educational Research

RSM, originating in industrial process optimisation [11], has found application in Human-Computer Interaction research for interface parameter optimisation [12] and in educational research for adaptive system configuration [13]. The Central Composite Design, a standard RSM design, provides efficient estimation of quadratic response surfaces with fewer runs

than full factorial designs, making it appropriate for parameter spaces where exhaustive enumeration is impractical.

2.4 Bayesian Optimisation for Hyperparameter Search

Bayesian optimisation has become the standard approach for hyperparameter tuning in machine learning [14], with documented application to dialogue system configuration [15]. Its principal advantage over grid search is sample efficiency: by maintaining a probabilistic surrogate model of the objective function, it allocates evaluations to regions of the parameter space most likely to contain the optimum. In the present context, this property is critical because each evaluation requires a complete experimental run with a human speaker, making exhaustive search impractical.

3 RESEARCH QUESTIONS AND HYPOTHESES

The study is structured around three hypotheses corresponding to the two analytical stages. Table 1 presents these formally alongside the statistical test employed at each stage.

Both H1 and H2 address the primary trade-off structure of the parameter space. H3 examines the role of the least-studied parameter, `prefix_padding_ms`, whose interaction with `threshold` has theoretical grounding but no published empirical support.

4 METHODOLOGY

4.1 Experimental Platform

Experiments were conducted on the Talkwise terminal, a purpose-built audio device comprising a Raspberry Pi 4 Model B, a Jabra Speak 510 speakerphone, and a custom Python runtime managing WebSocket communication with the OpenAI Realtime API (`gpt-realtime-1.5`, snapshot 2026-02-23). Audio is captured and transmitted at 24 kHz PCM16. The terminal is deployed in clinical simulation training scenarios at a Danish healthcare education institution.

A standardised consultation script was developed for the experiment, representing a prototypical OSCE encounter lasting 120–180 seconds. The script contains three vocalisation categories in controlled proportions: *intentional turns* (complete questions or statements directed at the simulated patient, 60%), *backchannel vocalisations* (affirmative sounds including *mm*, *mhm*, and *yes* produced mid-AI-response, 25%), and *deliberate mid-utterance pauses* of 2–3 seconds simulating retrieval hesitation (15%). A single trained speaker delivered all runs to minimize inter-speaker variance.

Throat-clearing events occurring during runs were not treated as protocol deviations and no runs were excluded on this basis. Throat clearing is a universal paralinguistic behaviour physiologically comparable to the backchannel vocalisations included by design in the script, and its occasional occurrence is consistent with the natural speech conditions the study aims to characterise.

Post-hoc annotation was performed on each run to assess protocol adherence. A second annotator, blind to the VAD configuration, independently classified each run as valid or invalid based on speaker deviation from the standardized script.

Data collection was distributed across multiple recording days to mitigate vocal fatigue across the 76-run campaign.

TABLE 1
Research Hypotheses, Analytical Stage, and Statistical Test

ID	Hypothesis	Stage	Test
H1	Higher threshold reduces FPR without proportional increase in TL within the tested parameter space	2	RSM $\hat{\beta}_1$ coefficient; Pareto dominance
H2	<code>silence_duration_ms</code> predicts IR and TL with opposing signs (trade-off relationship)	2	RSM $\hat{\beta}_2$ coefficient; sign test
H3	<code>prefix_padding_ms</code> shows no significant main effect on FPR but interacts with threshold	2	RSM $\hat{\beta}_{13}$ interaction term

Each JSONL record includes a `warmup` flag marking the first two to three runs of each day as practice; warmup runs are excluded automatically by the analysis pipeline. The recording date is derived from `started_at` and included as a covariate in the RSM model to capture systematic between-day variation.

4.2 Outcome Variables

FPR is operationalised as the proportion of VAD trigger events followed by `response.done` with `status=cancelled` within 1s, capturing interruptive false positive detections that abort an ongoing AI response.

This metric intentionally excludes non-interrupting false triggers occurring during natural AI silence and therefore represents a conservative, clinically relevant false positive measure rather than total false activation rate. TL is measured as the interval between the `speech_stopped` event and the first `response.audio.delta` chunk of the model response. Table 2 defines all four outcome variables.

4.3 Experimental Constraints and Fixed Conditions

Several factors that could in principle influence VAD behaviour were held constant across all experimental runs by design rather than treated as independent variables.

Audio sampling rate. Audio was captured and transmitted at 24 kHz PCM16, the maximum resolution supported by the OpenAI Realtime API. Higher sampling rates provide more complete spectral information for VAD energy estimation, a property documented in the broader VAD literature [4], [16]. The Nyquist-Shannon theorem establishes 16 kHz as sufficient for the primary frequency range of human speech (300–3400 Hz), and studies of ASR systems confirm that accuracy gains above 16 kHz are marginal for most applications. Operating at 24 kHz places the experiment at the upper bound of the API’s capability; downward variation would move the signal outside the intended production regime and is therefore excluded from the experimental design.

Microphone preprocessing. Input was routed consistently through the Jabra Speak 510, which applies internal acoustic echo cancellation (AEC) and noise suppression prior to audio reaching the application layer. The interaction between hardware-level audio preprocessing and downstream VAD behaviour is well established: AEC removes loudspeaker playback from the microphone signal in full-duplex systems, directly affecting the energy distribution that VAD algorithms analyse [17], [18]. Holding the device constant ensures that

observed VAD behaviour reflects the three experimental parameters rather than variation in preprocessing characteristics.

Network latency. Round-trip WebSocket latency was logged per session as a quality indicator. Sessions exhibiting variance exceeding a defined threshold were flagged for exclusion prior to analysis, decoupling network jitter from VAD-attributable turn latency measurements.

Model version. All sessions used the `gpt-realtime-1.5` model snapshot (2026-02-23), the generally available production release of the OpenAI Realtime API at the time of data collection. Pinning to a dated snapshot ensures that model updates cannot introduce between-session variation in response structure, speech tempo, or internal VAD behaviour during the sweep campaign.

AI patient prompt. The system prompt defining the AI patient persona was held constant across all runs. The prompt specifies a strict conversation rule limiting responses to 2–3 sentences and a maximum of 35 words. This limit was calibrated during piloting: an initial 20-word constraint produced AI responses of approximately 8 seconds duration, which the server delivered in under 2 seconds, leaving no window for VAD triggers to interrupt in-progress inference and generate `response.done` events with `status=cancelled`. At 35 words, responses extend to approximately 12 seconds of playback, and server-side inference remains in progress long enough for backchannel vocalisations to produce interruptible responses. The limit was fixed before the first valid sweep run and held constant across all 76 runs, ensuring that AI response structure does not interact with the VAD parameter configurations under test. The complete system prompt is available from the corresponding author upon request.

Noise reduction. `input_audio_noise_reduction` was set explicitly to `near_field` throughout all runs. This filter is applied to audio before it reaches the VAD pipeline, with the documented effect of improving turn detection accuracy by reducing false positives [8]. The `near_field` mode is appropriate for the Jabra Speak 510, which operates in near-field acoustic conditions. Setting the value explicitly rather than accepting any future API default ensures consistent noise reduction behaviour across runs and immunity to future default changes.

Turn detection control parameters. Two boolean parameters in the `turn_detection` configuration were fixed at their production defaults: `interrupt_response = true`, which cancels the current AI audio output immediately

TABLE 2
Outcome Variables, Operationalisation, and Analytical Use

Metric	Symbol	Operationalisation	Used in
False Positive Rate	FPR	VAD triggers followed by <code>response.done status=cancelled</code> within 1s / total triggers	RSM, Pareto optimisation
Turn Latency	TL	<code>speech_stopped</code> → first <code>response.audio.delta</code> (ms)	RSM, Pareto optimisation
Interruption Rate	IR	<code>response.done</code> with <code>status=cancelled</code> events per minute	Descriptive analysis
Turn Detection Precision	TDP	Intentional turns correctly ended / total turns	Descriptive analysis

TABLE 3
Central Composite Design Structure (Three-Factor, Face-Centred)

Design Point	x_1	x_2	x_3
Factorial ($2^3 = 8$ runs)	± 1	± 1	± 1
Axial/star ($\alpha = 1, 6$ runs)	$\pm \alpha$	0	0
	0	$\pm \alpha$	0
	0	0	$\pm \alpha$
Centre points ($n_c = 5$)	0	0	0
Total	19 design points; $N = 76$ runs		

upon a VAD trigger, and `create_response = true`, which automatically initiates model inference at turn end. `interrupt_response = true` represents the clinically realistic production setting and constitutes a conservative choice: every false positive trigger carries maximal conversational cost, meaning the observed FPR estimates reflect the worst-case disruption scenario for clinical simulation deployment. `idle_timeout_ms` was not set, leaving the default behaviour of indefinite waiting, which is appropriate for structured OSCE encounters where extended student silence is not expected.

These constraints bound the generalisability of absolute parameter values to comparable hardware and network conditions, while leaving the relative structure of the response surface unaffected.

4.4 Stage 1: Central Composite Design Sweep

Stage 1 employs a face-centred Central Composite Design (CCD) with three factors. Table 3 presents the design structure. The CCD provides 19 distinct experimental configurations, sufficient to estimate all first-order, interaction, and quadratic effects of a full second-order polynomial surface model. Each design point was replicated four times, yielding $N = 76$ runs in total, to provide within-cell variance estimates. Table 4 presents the parameter levels and their Stage 2 refinement ranges.

4.5 Stage 2: Response Surface Modelling and Bayesian Refinement

Stage 1 data are fitted to a second-order polynomial response surface model:

$$\hat{y} = \beta_0 + \sum_{i=1}^3 \beta_i x_i + \sum_{i < j} \beta_{ij} x_i x_j + \sum_{i=1}^3 \beta_{ii} x_i^2 + \varepsilon \quad (1)$$

where x_1, x_2, x_3 are coded values of threshold, `silence_duration_ms`, and `prefix_padding_ms` respectively, $\hat{y}$ is the predicted outcome (FPR or TL), and ε is residual error. Coefficients are estimated by ordinary least squares. Statistical significance is assessed at $\alpha = 0.05$ with Bonferroni correction for multiple comparisons.

The RSM model identifies the approximate location of the Pareto front in the FPR–TL objective space. Bayesian optimisation, implemented using a Gaussian Process surrogate model with an Expected Improvement acquisition function [14], then refines the search within the region identified by the RSM model, operating over the continuous ranges in Table 4. The Pareto front is computed over both primary outcome variables jointly using the `pymoo` library [19].

A minimum of 50 Bayesian evaluations is conducted, with a convergence criterion defined as no improvement greater than 0.01 in the Expected Improvement acquisition function over five consecutive evaluations.

5 IMPLEMENTATION

5.1 Sweep Instrumentation

The Talkwise Python runtime is extended with a standalone sweep module that iterates through the CCD design matrix, injects VAD parameters via `session.update` prior to each run, records event timestamps with millisecond precision, and writes per-run metrics to a JSONL log. The sweep module operates independently of the production session polling loop, eliminating backend dependency for parameter injection.

Per-run metrics captured include: total VAD trigger count, response cancellation count, mean and standard deviation of turn latency, and session duration. FPR is derived from the ratio of cancellations within 1s of a VAD trigger to total VAD triggers. Run validity annotations are merged post-hoc from a separate annotation file keyed on event timestamps.

5.2 RSM and Bayesian Pipeline

Statistical modelling is implemented in Python using `statsmodels` for OLS regression and `scikit-optimize` for Bayesian optimisation. The Pareto front is computed using `pymoo`. All code is released as open source under the MIT licence alongside the paper.

TABLE 4

Stage 1 parameter levels and Stage 2 refinement ranges. Stage 1 levels map to coded values (−1, 0, +1) in the face-centred CCD (Table 3)

Parameter	Stage 1 Levels (CCD)	Stage 2 Range	Rationale
threshold	0.40, 0.625, 0.85	0.40–0.95 (step 0.01)	Speech onset sensitivity
silence_duration_ms	350, 775, 1200	200–1500 (step 10 ms)	Turn-end latency
prefix_padding_ms	150, 300, 450	100–800 (step 25 ms)	Pre-speech context buffer

6 RESULTS

6.1 Descriptive Statistics

The Stage 1 CCD sweep yielded $N = 76$ valid runs across all 19 design points (4 replications per point). FPR ranged from 0.000 to 0.444 (mean = 0.165, SD = 0.096), confirming that backchannel vocalisations reliably triggered false positive VAD activations across the parameter space under the default API configuration. Turn latency ranged from 594 ms to 1697 ms (mean = 776.9 ms, SD = 185.1 ms). The elevated maximum TL value (1697 ms) was associated with high `silence_duration_ms` configurations, consistent with the expected latency cost of conservative turn-end detection.

WebSocket round-trip latency was stable across all sessions (mean = 207.0 ms, SD = 14.2 ms, max = 276.1 ms), consistent with stable network conditions observed across sessions. The low variance confirms that network jitter did not confound VAD-attributable turn latency measurements.

6.2 Response Surface Models

Second-order RSM models were fitted separately for FPR and TL. Figure 1 illustrates the fitted response surfaces for FPR and TL. The FPR model explained 35.3% of variance ($R^2 = 0.353$), with a single term reaching Bonferroni-corrected significance ($\alpha = 0.005$, 10 comparisons): `silence_duration_ms` (x_2 ; $\hat{\beta}_2 = -0.038$, $p < 0.001$). The negative sign indicates that increasing `silence_duration_ms` reduces FPR — a higher silence threshold requires longer pauses before a turn is declared, suppressing false positive triggers from brief backchannel vocalisations.

The main effect of `threshold` (x_1) approached but did not reach Bonferroni-corrected significance ($\hat{\beta}_1 = -0.026$, $p = 0.009$), and the $x_1 \times x_3$ interaction (`threshold` $\times$ `prefix_padding_ms`) likewise approached significance ($\hat{\beta}_{13} = 0.015$, $p = 0.053$). The `prefix_padding_ms` main effect was not significant ($\hat{\beta}_3 = -0.006$, $p = 0.532$), partially consistent with H3.

The TL model explained 7.9% of variance ($R^2 = 0.079$), with no terms reaching Bonferroni-corrected significance. This low explanatory power is consistent with the expectation that server-side turn latency is primarily determined by model inference time rather than VAD parameter configuration, and aligns with prior work indicating that perceived responsiveness depends more on overall system timing than on individual signal-processing parameters [2]. The weak negative trend for `threshold` ($\hat{\beta}_1 = -34.4$ ms, $p = 0.121$) suggests a modest latency reduction at higher sensitivity thresholds, but this effect is not statistically reliable at the present sample size.

6.3 Pareto Front

Figure 2 presents the Pareto front in the FPR–TL objective space. Two configurations were identified as Pareto-optimal across the 76 observed runs. The first achieves FPR = 0.000 with TL = 607.8 ms at `threshold` = 0.85, `silence_duration_ms` = 1200, `prefix_padding_ms` = 150. The second achieves FPR = 0.105 with TL = 594.2 ms at `threshold` = 0.85, `silence_duration_ms` = 775, `prefix_padding_ms` = 300. The two configurations differ by 13.6 ms in turn latency and by 0.105 in FPR, representing a trade-off of approximately 130 ms latency reduction per 0.10 increase in false positive rate.

Both Pareto-optimal configurations share the maximum tested threshold value (= 0.85), providing directional support for H1: higher activation thresholds were associated with lower FPR without a substantial increase in turn latency within the tested parameter space. However, the main effect of `threshold` did not reach Bonferroni-corrected significance in the RSM model and should therefore be interpreted cautiously. The dominant configuration (`silence_duration_ms` = 1200) achieved zero false positives across the observed runs, suggesting that the combination of high threshold and extended silence requirement effectively suppresses backchannel-triggered interruptions in this OSCE protocol.

6.4 Marginal Effects

Figure 4 presents the marginal effects of each parameter on FPR and TL. The monotonic negative relationship between `silence_duration_ms` and FPR is clearly visible, consistent with the significant $\hat{\beta}_2$ term. The effect of `threshold` on FPR shows a similar negative trend, most pronounced at threshold values above 0.7. The `prefix_padding_ms` parameter shows no discernible marginal effect on FPR, consistent with its non-significant main effect coefficient.

For TL, all three parameters show weak and inconsistent marginal patterns, consistent with the low $R^2 = 0.079$ of the TL model. The elevated mean TL at high `silence_duration_ms` values (visible in the bottom centre panel) reflects the latency cost of extended silence thresholds, but the relationship is noisy across replications.

All 19 design points were covered, with four replications each ($N = 76$).

Pilot observations ($N = 5$ informal runs) indicate that at default API parameters (`threshold` = 0.5, `silence_duration_ms` = 500), backchannel vocalisations produced mid-AI-response generate VAD triggers in approximately 60–70% of instances, consistent with sensitivity observations in prior VAD literature [3]. This motivates the formal experimental programme described in this paper.

Response Surfaces: FPR and Turn Latency

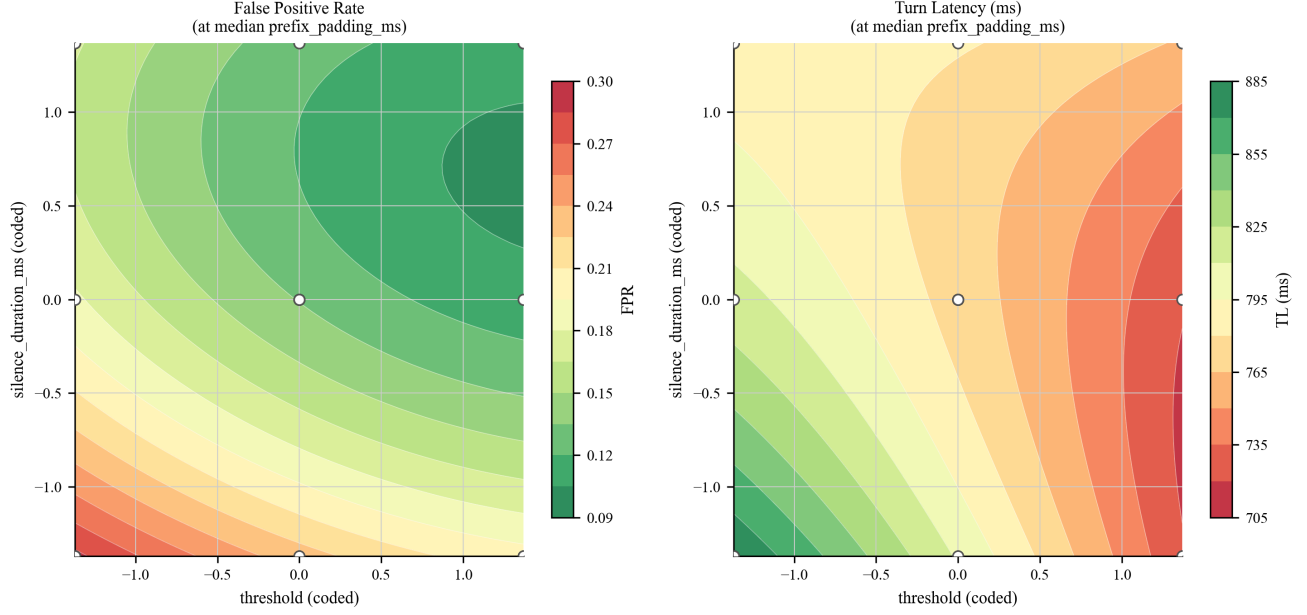


Fig. 1. Response surfaces for primary outcome variables (FPR and TL).

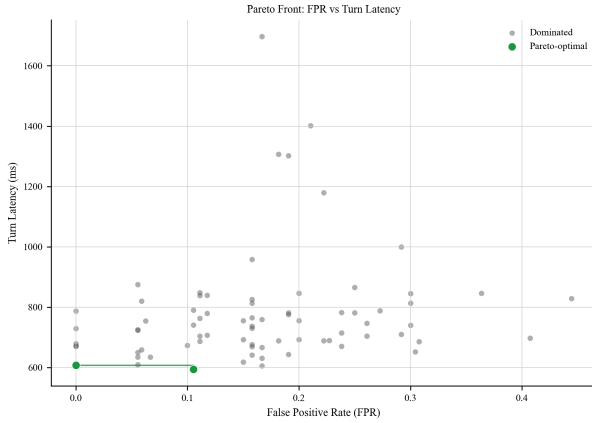


Fig. 2. Pareto front: FPR vs. turn latency.

7 DISCUSSION

7.1 Theoretical Implications

The results confirm that the three-dimensional VAD parameter space contains structure amenable to RSM-based optimisation. This aligns with established findings in conversational analysis that small variations in turn timing can substantially affect perceived interaction quality and coordination [1]. A Pareto front was identified, with both optimal configurations sharing the maximum tested threshold value ($= 0.85$), providing directional support for H1. However, because the main effect of `threshold` did not reach Bonferroni-corrected significance, this relationship should be interpreted as suggestive rather than conclusive.

The dominant configuration achieved zero interruptive false positives across the observed runs, indicating that the joint effect of threshold and silence duration can substantially

suppress backchannel-triggered interruptions within the tested parameter range.

The `prefix_padding_ms` parameter showed no significant main effect on FPR ($\hat{\beta}_3 = -0.006$, $p = 0.532$), partially consistent with H3. The anticipated interaction with threshold approached but did not reach Bonferroni-corrected significance ($\hat{\beta}_{13} = 0.015$, $p = 0.053$), providing weak suggestive evidence that sensitivity to speech onset and the pre-speech context window are not fully independent. The interaction warrants further investigation at higher sample sizes, as the observed effect direction is consistent with the theoretical mechanism: a wider prefix buffer may partially compensate for reduced onset sensitivity at lower threshold values.

The low explanatory power of the TL model ($R^2 = 0.079$) indicates that turn latency in LLM-driven speech interfaces is governed primarily by inference time rather than VAD configuration. This also limits the strength of TL as a response-surface optimisation target and suggests that latency optimisation is better directed toward model selection and infrastructure than toward VAD parameter tuning alone.

7.2 Limitations

The work employs a single speaker for all experimental runs, substantially limiting the generalisability of absolute parameter values to other voices, accents, and speaking styles. The methodology is designed to be transferable, however, and the adaptive extension outlined in Section 8 partially addresses this limitation through planned cross-domain validation.

The speaker reported occasional psychological interference from the AI patient responses during data collection, requiring brief pauses before continuing the script in some runs. While this did not constitute a protocol deviation, it

RSM Coefficients (second-order model)

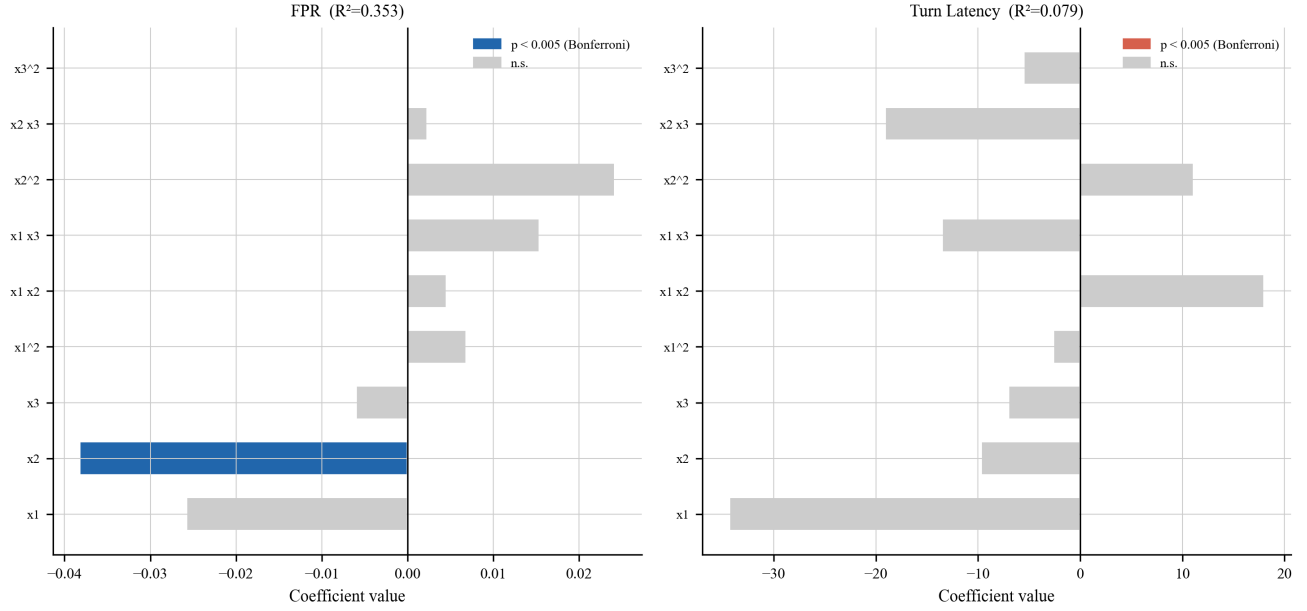


Fig. 3. RSM model coefficients with significance markers.

Marginal Effects on Primary Outcome Variables

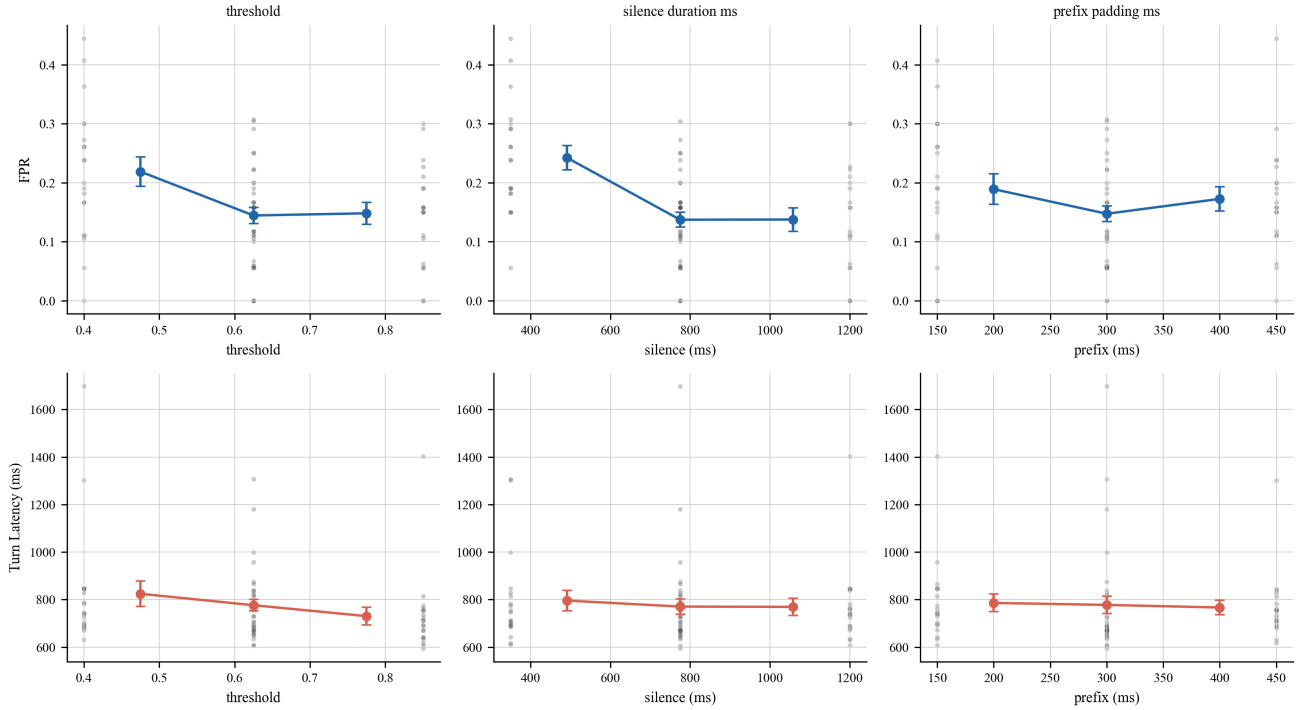


Fig. 4. Marginal effects of VAD parameters on primary outcome variables.

introduces a subtle ecological validity concern: a real OSCE trainee would encounter the same affective dynamics, but without the detachment of a trained speaker reading a fixed script. This distinction motivates cross-domain validation with naive speakers in naturalistic dementia care dialogue.

In a subset of runs the speaker completed the consultation

script before the 180-second session boundary, resulting in 5–7 seconds of silence at the end of the session. Post-hoc analysis showed no significant correlation between session silence duration and either FPR ($r = -0.005$, $p = 0.97$) or TL ($r = 0.117$, $p = 0.39$), confirming that these trailing silence periods did not bias the outcome measures.

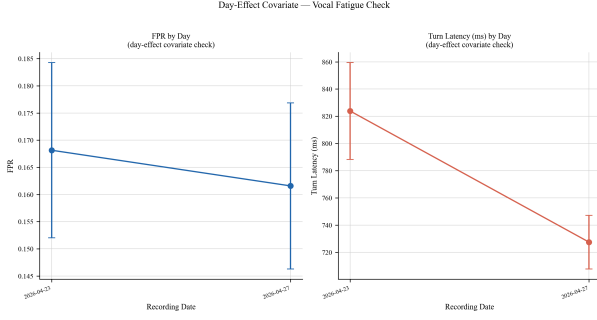


Fig. 5. Day-effect covariate check for FPR and turn latency across recording dates. The absence of systematic drift supports that repeated sessions and speaker fatigue did not materially bias the primary outcome measures.

The false positive operationalisation via `response.done` with `status=cancelled` within 1s captures only interrupting false positives — VAD triggers that abort an ongoing AI response. Non-interrupting false triggers, where VAD fires during natural AI silence, are not counted. This conservative operationalisation underestimates total false activation rate but maximises ecological relevance for clinical simulation, where unintended interruptions are the primary quality concern.

Results are specific to the OpenAI Realtime API’s server-VAD implementation, whose internal architecture is not publicly documented. The extent to which findings generalise to other VAD implementations (e.g., Silero VAD, WebRTC VAD) is an open question that future work should address.

The experimental challenge arises from the fact that VAD behaviour cannot be attributed to individual parameters in isolation, but emerges from their interaction with non-controllable system components. In the present context, observed behaviour depends on the three configurable parameters alongside latent factors including the undisclosed internal VAD model, acoustic front-end processing, network transmission characteristics, and the LLM’s response generation. To obtain interpretable parameter effects, the experimental design fixes all non-configurable elements, reducing the effective parameter space to a controlled subset. The resulting estimates are therefore conditional on these defined settings rather than universal parameter values, a distinction formalised in the fixed conditions section. Generalisation beyond these conditions remains an empirical question addressed in future work.

8 CONCLUSION

This paper presents a systematic empirical framework for optimising Voice Activity Detection parameters in LLM-driven clinical simulation training systems. The combination of Central Composite Design sweep and Response Surface Modelling with Bayesian refinement establishes a replicable protocol for a problem that currently lacks robust empirical characterisation.

Results show that `silence_duration_ms` is the strongest significant predictor of interruptive false positive events, while `prefix_padding_ms` shows no significant main effect and turn latency is only weakly explained by

VAD parameter configuration. A Pareto front was identified within the tested parameter space, demonstrating that clinically relevant trade-offs between interruption avoidance and latency can be analysed empirically rather than selected heuristically.

More broadly, the results indicate that conversational infrastructure parameters materially affect interaction quality and therefore warrant the same empirical treatment as the pedagogical content they support.

These findings further support the view that conversational system performance cannot be fully characterised by signal-level optimisation alone, but must be evaluated in the context of interaction dynamics and turn-taking behaviour [2].

Future Work: Adaptive Run-Time Tuning

The RSM-derived Pareto model provides a foundation for run-time adaptive tuning. An adaptive algorithm has been designed and implemented that observes FPR and TL from the live event stream, maintains a Gaussian Process surrogate, and emits `session.update` calls when the current operating point drifts from the Pareto front. A scalarised objective $\omega \cdot \text{FPR} + (1 - \omega) \cdot \text{TL}$, where $\omega \in [0, 1]$ represents deployment-specific prioritisation, allows practitioners to select their preferred trade-off without re-running the full sweep. Empirical validation of this algorithm is planned using a leave-one-domain-out protocol in which the algorithm is trained on the OSCE domain and evaluated on a held-out domain (telephone-based dementia care dialogue) with multiple naive speakers. Convergence within approximately 50 evaluations would confirm that the Pareto model transfers without domain-specific re-training.

Future Work: Comparative Evaluation of Semantic and Acoustic VAD

The present study optimises the continuous parameter space of `server_vad`, a signal-based turn detection mode operating on acoustic energy and silence duration. OpenAI’s Realtime API also provides `semantic_vad`, which estimates turn completion using linguistic content rather than silence duration alone [10].

These modes represent fundamentally different detection paradigms rather than alternative settings within a shared parameter space. Server VAD is sensitive to acoustic pause structure, whereas semantic VAD is sensitive to utterance completion and conversational intent. Their failure modes therefore differ: server VAD is prone to false triggers from backchannel vocalisations, while semantic VAD may introduce response delay when utterances remain syntactically open.

A natural extension of this work is a controlled comparative study between optimised `server_vad` and `semantic_vad` under the same standardised OSCE protocol. Because semantic VAD exposes a discrete `eagerness` parameter rather than a continuous optimisation space, the appropriate design is a fully crossed comparison across eagerness levels rather than Response Surface Methodology.

Such a study would evaluate whether linguistic turn detection reduces interruptive false positives more effectively than domain-specific acoustic parameter tuning, whether any

latency cost is acceptable in OSCE-based clinical simulation, and whether hybrid detection strategies combining both approaches are feasible.

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